Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity

Beniaiche Ahmed[1,*], Larabi Abderrahim[1], Belkadi Mbarek[2]

[1]Laboratory of Fluid Mechanics, Ecole Militaire Polytechnique, Algiers, Algeria
[2]Laboratory of Turbomachinery, Ecole Militaire Polytechnique, Algiers, Algeria

Abstract

The spatial and temporal organization of wall actuation determines how mechanical input is distributed across the flow structures of a driven cavity, yet the resulting viscous-dissipation cost cannot be inferred from the actuation spectrum or from the time-mean velocity field alone. This study examines the relationship between wall-actuation structure, hydrodynamic response, and cycle-mean viscous dissipation in a two-dimensional lid-driven cavity. Stationary harmonic profiles and three physics-informed actuation branches are transferred to an independent multiple-relaxation-time lattice-Boltzmann solver and compared at a common quadratic actuation norm. For stationary controls at $Re = 500$, the first spatial harmonic reduces dissipation by 17.7% relative to a corner-regularized uniform-lid reference, whereas the second harmonic and the mean profile of the second-mode-dominated branch increase it by 36.8% and 44.3%, respectively. Grid-refinement and local strain-rate analyses associate this ordering with the redistribution of dissipation between corner shear layers and the cavity interior. For time-dependent branches, the cycle-mean dissipation differs markedly from the dissipation of the time-mean flow. In particular, the temporally dominated branch has a cycle-mean dissipation approximately 3.1 times that of the predominantly stationary branch, despite its much weaker mean-flow contribution. The results demonstrate that the energetic cost of wall actuation depends on both the spatial organization and the temporal fluctuations of the resulting flow, and that physical performance must be assessed using cycle-averaged dissipation rather than modal accessibility or mean-field diagnostics alone.

Keywords: lid-driven cavity; wall actuation; viscous dissipation; energy budget; mean-flow diagnostics; mode selection

*Corresponding author: beniaiche_ahmed25@yahoo.fr

Nomenclature

Roman symbols	
A_ε	half-spread of $D(t)$ over the cycle, i.e. the variability concealed by a single cycle-mean number
AR	cavity aspect ratio, L_x/L_y
C_3	spectral rank concentration, sum of the three largest control-energy fractions
D	instantaneous viscous dissipation rate, $(2/Re)\int_\Omega \mathbf{S}:\mathbf{S}\,d\Omega$
$\overline{D}$	cycle-mean dissipation, $\langle D\rangle_t$
E_{act}	instantaneous actuation energy of the lid profile, $\int_0^1 U_{\mathrm{lid}}^2\,dx$
E_r	realized cycle-averaged actuation energy of a transferred control
E^*	prescribed target actuation energy
f_1, f_2, f_t	first-harmonic, second-harmonic, and explicitly time-dependent control-energy fractions
H_N	normalized branch-selection entropy
K	kinetic energy, $\frac{1}{2}\int_\Omega \lvert\mathbf{u}\rvert^2\,d\Omega$
L, L_x, L_y	cavity side, streamwise and wall-normal lengths
N	number of lattice nodes per cavity side (LBM resolution)
P_{lid}	mechanical power injected through the lid, $\int_0^1 U_{\mathrm{lid}}\tau_w\,dx$
R_K	ratio of fluctuating to mean kinetic energy
Re	Reynolds number, $U_{\mathrm{ref}}L/\nu$
$\mathbf{S}$	strain-rate tensor, $(\nabla\mathbf{u}+\nabla\mathbf{u}^{\mathsf{T}})/2$
t, T	time and forcing period
$\mathbf{u}=(u,v)$	velocity vector and its streamwise/wall-normal components
u'	fluctuating velocity, $\mathbf{u}-\langle\mathbf{u}\rangle$
u'_{rms}	root-mean-square temporal fluctuation at a fixed point
U_{lid}	lid velocity profile (the actuation)
U_{ref}	reference velocity, the amplitude of the uniform lid
x, y	streamwise and wall-normal coordinates
Greek symbols and branch symbols	
B_1, B_2, B_3	empirical branch classes (second-harmonic dominated, time-dependent dominated, residual)
B_{first}	first-harmonic branch class of the full-spectrum audit
δ	corner taper width
ε	integrated viscous dissipation accumulated over an LBM sampling window
λ_{var}	weight of the variance-regularization term of the training loss
ν	kinematic viscosity
τ_w	wall shear stress on the moving lid
ψ	stream function
Ω	cavity domain
Abbreviations	
GCI	grid-convergence index
LBM	lattice-Boltzmann method
LOCO	leave-one-Reynolds-condition-out (cross-validation)
MRT	multiple-relaxation-time
PINN	physics-informed neural network
RMSE	root-mean-square error

1 Introduction

Wall actuation is one of the few ways in which the vortical structure of a confined recirculating flow can be modified without changing its geometry. In a lid-driven cavity the entire vortex topology is generated by the tangential motion of a single boundary, so any spatial structure imposed on that boundary acts directly on the mechanism that sustains the flow. The cavity is therefore a natural configuration in which to ask a question that is central to flow control but rarely posed in energetic terms: when two different actuations require the same control effort, which one produces the less dissipative flow, and why? The canonical cavity solution and its benchmarks are well established [1, 2, 3, 4, 5], and the flow has long served as a test bed for control strategies [6, 7], yet its dissipation budget under distributed wall actuation has received comparatively little attention.

The energetic mechanism is twofold, and the two parts can act in opposite directions. Changing the wall velocity changes the mechanical power injected into the fluid, $P_{\mathrm{lid}} = \int_0^1 U_{\mathrm{lid}} \tau_w \, dx$, where τ_w is the wall shear stress; it also changes the spatial distribution of the strain-rate tensor $\mathbf{S}$, and with it the partition of viscous dissipation between the lid shear layer, the side-wall layers, the cavity interior and the two upper corners, where a discontinuous lid produces a stress singularity. A profile can therefore reduce the peak strain at the corners while increasing the strain everywhere else, and the net effect is not predictable from the shape of the actuation alone. The distinction matters because the two natural comparison protocols answer different questions. Comparing profiles at the same lid-velocity scale [32] measures what a given actuator amplitude achieves; comparing them at the same quadratic actuation norm $E_{\mathrm{act}} = \int_0^1 U_{\mathrm{lid}}^2 \, dx$ measures what a given control effort achieves. The two rankings need not agree, and only the second is a statement about the energetic response at fixed actuation norm. A further distinction, specific to time-dependent actuation, is that the dissipation of the time-mean field is not the time mean of the instantaneous dissipation: because D is quadratic in the velocity gradients, cycle-averaged dissipation always exceeds the dissipation of the mean flow, and the gap can be orders of magnitude when the fluctuations dominate [23, 24].

Spatially structured and time-periodic actuations can also support more than one hydrodynamic response. A periodic boundary-value problem of this type admits several coexisting branches, and which one is reached depends on the formulation, the regularization and the initialization of the identification procedure. A preceding study by Beniaiche [32] investigated the identification and independent numerical verification of conditional modal collapse in physics-informed active control of the lid-driven cavity. It established that the identified actuation is not characterized by a unique, universally selected modal structure: predominantly stationary second-harmonic controls and strongly time-dependent alternatives can be recovered depending on initialization, regularization, and control parameterization. Independent MRT-LBM simulations further demonstrated that these identified controls can produce physically distinct cavity responses. The emphasis of that work was on branch identification, conditional accessibility, and independent numerical reproducibility. The present study addresses the complementary fluid-mechanical question of how these spatial and temporal actuation structures affect the viscous-dissipation budget of the resulting flow. Rather than treating modal recovery as a performance criterion, we quantify the dissipation of stationary and periodic responses under a common quadratic actuation norm, distinguish the dissipation of the time-mean field from the cycle-mean dissipation, and identify the strain-rate distributions responsible for the observed energetic differences.

The question addressed in this paper is accordingly:

> How does the spatio-temporal structure of wall actuation determine the cycle-mean viscous dissipation of the resulting cavity flow at a prescribed actuation-energy level?

The sequence of the paper follows that question. Section 2 fixes the physical and energetic framework, and in particular separates the quadratic actuation norm from the mechanical power transmitted through the wall. Section 3 defines the stationary modes and the three identified branches that are compared. Section 4 describes how the actuations are identified and how the flows are computed independently. Section 5 reports the energetic response of the stationary modes, Section 6 the cycle-mean response of the time-dependent branches, and Section 7 interprets both in terms of local strain and fluctuating gradients. Section 8 concludes.

2 Physical formulation and energetic framework

2.1 Controlled cavity and governing equations

We consider a two-dimensional incompressible flow in the square cavity $0 \le x \le 1$, $0 \le y \le 1$, made dimensionless with the cavity height L and a reference velocity U_0 (Fig. 1a). The lower and

lateral walls are no-slip and stationary; the upper wall moves tangentially with a velocity profile $U_{\mathrm{lid}}(x,t)$ that is the control variable. The velocity field $\mathbf{u}=(u,v)$ and the pressure p satisfy

$$\frac{\partial \mathbf{u}}{\partial t}+(\mathbf{u}\cdot\nabla)\mathbf{u}=-\nabla p+\frac{1}{Re}\nabla^2\mathbf{u}, \qquad \nabla\cdot\mathbf{u}=0, \tag{1}$$

with $Re=U_0L/\nu$, subject to

$$\mathbf{u}=\mathbf{0} \ \text{ on } x=0,\ x=1,\ y=0, \qquad u=U_{\mathrm{lid}}(x,t), \quad v=0 \ \text{ on } y=1. \tag{2}$$

A pointwise tangential velocity along the lid idealizes an array of independently driven wall segments, or a distributed momentum source such as a dielectric-barrier-discharge actuator, whose near-wall forcing is equivalent in the time-averaged sense used here to a tangential wall velocity [13]. Because U_0 is a scaling velocity and not a property of the lid, a profile may change sign along x or in time; the amplitude of the actuation is fixed separately by the norm constraint introduced in Sec. 2.2.

2.2 The quadratic actuation norm and the mechanical wall power

Two different scalar quantities characterize a wall actuation, and they must not be conflated. The first is the *quadratic actuation norm*

$$E_{\mathrm{act}}(t)=\int_0^1 U_{\mathrm{lid}}^2(x,t)\,dx, \qquad \overline{E}_{\mathrm{act}}=\frac{1}{T}\int_0^T E_{\mathrm{act}}(t)\,dt, \tag{3}$$

which measures the squared amplitude of the commanded wall velocity, has units of U_0^2L, and is the quantity held fixed in all comparisons below. It is a norm of the boundary data, not a work: it contains no information about the fluid response. The second is the *mechanical power* transmitted by the lid,

$$P_{\mathrm{lid}}(t)=\int_0^1 U_{\mathrm{lid}}(x,t)\,\tau_w(x,t)\,dx, \qquad \tau_w=\frac{1}{Re}\left.\frac{\partial u}{\partial y}\right|_{y=1}, \tag{4}$$

positive when the wall does work on the fluid. P_{lid} depends on the actuation and on the flow through the wall shear stress, and it is the quantity that enters the energy balance. Two actuations of equal $\overline{E}_{\mathrm{act}}$ can therefore inject very different amounts of power; conversely, a large norm does not guarantee a large power if the wall shear is weak.

Holding $\overline{E}_{\mathrm{act}}$ fixed also fixes the root-mean-square wall velocity, $\overline{E}_{\mathrm{act}}^{1/2}=0.5$ at the value $E^*=0.25$ used throughout, but not the maximum wall velocity. A uniform lid of unit shape then has amplitude 0.5, whereas a first harmonic of the same norm has amplitude $\sqrt{2E^*}=0.7071$; a peakier profile has a larger maximum still. Because the viscosity is fixed, a profile whose peak wall velocity is $U_{\max}$ realizes an effective Reynolds number $Re_{\max}=U_{\max}Re$ based on that peak. The comparisons below are therefore comparisons at equal quadratic actuation norm, at fixed viscosity and geometry, and *not* at equal $Re_{\max}$; the values of $U_{\max}$ and $Re_{\max}$ are tabulated for every case (Tables 1 and 2) so that the distinction is explicit at every step. This convention differs from the fixed-lid-velocity comparison of Ref. [32], in which the peak velocity was the same for all profiles and the norm differed by up to a factor of four.

2.3 Viscous dissipation and its cycle mean

For an incompressible Newtonian fluid the instantaneous viscous dissipation rate is

$$D(t)=\frac{2}{Re}\int_\Omega \mathbf{S}:\mathbf{S}\,d\Omega, \qquad \mathbf{S}=\tfrac{1}{2}\left(\nabla\mathbf{u}+\nabla\mathbf{u}^T\right), \tag{5}$$

and the quantity relevant to a periodic actuation is its cycle average

$$\overline{D}=\frac{1}{T}\int_0^T D(t)\,dt. \tag{6}$$

Splitting the velocity into a time mean and a fluctuation, $\mathbf{u} = \overline{\mathbf{u}} + \mathbf{u}'$ with $\mathbf{S} = \overline{\mathbf{S}} + \mathbf{S}'$, the cycle mean decomposes exactly as

$$\overline{D} = \underbrace{\frac{2}{Re}\int_\Omega \overline{\mathbf{S}} : \overline{\mathbf{S}}\, d\Omega}_{D(\overline{\mathbf{u}})} + \underbrace{\frac{2}{Re}\left\langle \int_\Omega \mathbf{S}' : \mathbf{S}'\, d\Omega \right\rangle}_{\text{fluctuating part}}, \tag{7}$$

because the cross term vanishes upon averaging. The two terms coincide only in the steady limit; otherwise $\overline{D} > D(\overline{\mathbf{u}})$, with a gap that measures how much of the dissipation is carried by the unsteady part of the response. Evaluating D on a time-averaged field is thus not an approximation of $\overline{D}$ with a small error: it is a different functional, and for a strongly unsteady flow it can invert the ranking of two actuations.

2.4 Mechanical energy budget

With the kinetic energy $K = \frac{1}{2}\int_\Omega |\mathbf{u}|^2\, d\Omega$, the momentum equation gives

$$\frac{dK}{dt} = P_{\text{lid}}(t) - D(t). \tag{8}$$

For a periodic state the left-hand side integrates to zero over one period, so

$$\boxed{\overline{P}_{\text{lid}} = \overline{D}} \tag{9}$$

is an exact identity of the governing equations. In the numerical evaluation of Eq. (9) both terms are computed from the same discrete fields, but P_{lid} involves a wall-normal derivative at the boundary, which is evaluated with a one-sided finite difference and therefore carries a truncation error that D does not. The budget is used below as a verification of the transfer and of the periodicity of the response, with that estimator uncertainty made explicit (Sec. 6.4); it is not used as an exact closure constraint.

3 Actuation modes and hydrodynamic states

3.1 Stationary reference profiles

Five stationary profiles define the comparison, all normalized to $E^* = 0.25$. The *uniform* lid $U_{\text{lid}} = 0.5$ is the reference of the problem statement; it carries a tangential-velocity discontinuity at the two upper corners, which is retained because it is the classical boundary condition, but which is shown in Sec. 5.2 to prevent the convergence of its own dissipation. A *corner-regularized uniform* lid, constructed with an initial plateau $U_p = 0.5$ and a raised-cosine taper of width $\delta = 0.05$ at each corner

$$U_{\text{lid}}^{\text{raw}}(x) = \begin{cases} \dfrac{U_p}{2}\left[1 - \cos\left(\dfrac{\pi x}{\delta}\right)\right], & 0 \le x < \delta, \\ U_p, & \delta \le x \le 1-\delta, \\ \dfrac{U_p}{2}\left[1 - \cos\left(\dfrac{\pi(1-x)}{\delta}\right)\right], & 1-\delta < x \le 1, \end{cases} \tag{10}$$

then rescaled to the prescribed norm:

$$U_{\text{lid}}(x) = c\, U_{\text{lid}}^{\text{raw}}(x), \qquad c = \sqrt{\frac{E^*}{\displaystyle\int_0^1 (U_{\text{lid}}^{\text{raw}})^2\, dx}}. \tag{11}$$

For $\delta = 0.05$, the raw profile carries $E_{\text{raw}}/E_{\text{uniform}} = 1 - 5\delta/4 = 0.9375$, a 6.25% loss which is restored by the rescaling factor $c = \sqrt{16/15} \approx 1.033$; the final plateau value is approximately

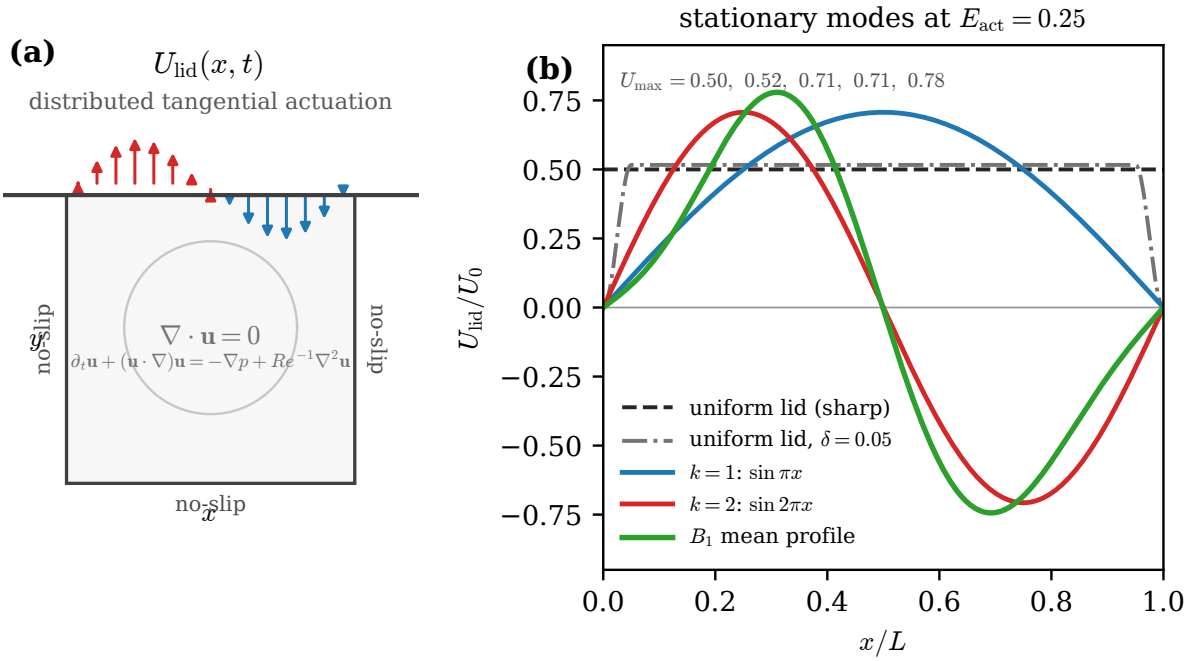


Figure 1: (a) Geometry of the controlled cavity. The upper wall moves tangentially with a distributed profile $U_{\mathrm{lid}}(x,t)$; the three other walls are no-slip and stationary. (b) The five stationary actuation profiles compared in this study, all normalized to the same quadratic actuation norm $E^*=\int_0^1 U_{\mathrm{lid}}^2\,dx=0.25$. The maximum wall velocity differs between profiles at equal norm: $U_{\max}=0.50$ for the uniform lid, 0.71 for the two harmonics and 0.78 for the B_1 mean profile.

0.5164, corresponding to $U_{\max}=0.52$ in table 1. This regularized profile removes the discontinuity and provides a converged reference. The first and second spatial harmonics, $U_{\mathrm{lid}}=\sqrt{2E^*}\sin(\pi x)$ and $\sqrt{2E^*}\sin(2\pi x)$, are the two elementary spatially structured controls. Finally, the time-mean profile of the identified B_1 branch (Sec. 3.2) is included as the stationary approximation of a physics-informed control.

3.2 Identified actuation branches

The time-dependent actuations are three representatives identified by the physics-informed procedure of Sec. 4 and classified by the modal energy of their control coefficients (details of the identification campaign and classification rules are provided in Supplementary Secs. S1–S3, particularly Sec. S2.3, Tables S2–S3 and Fig. S1). Writing the lid as

$$U_{\mathrm{lid}}(x,t)=\sum_{m=1}^{N_x}\sum_{n=0}^{N_t-1}a_{mn}\sin(m\pi x)\cos\left(\frac{2n\pi t}{T}\right), \tag{12}$$

with forcing period $T=2$, the energy of coefficient (m,n) is $\mathcal{E}_{m0}=\frac{1}{2}a_{m0}^2$ and $\mathcal{E}_{mn}=\frac{1}{4}a_{mn}^2$ for $n\geq 1$, with normalized fractions $f_{mn}=\mathcal{E}_{mn}/\sum\mathcal{E}_{ij}$. The stationary second-harmonic fraction is $f_2\equiv f_{20}$ and the total time-dependent fraction is $f_t=\sum_{m=1}^{N_x}\sum_{n=1}^{N_t-1}f_{mn}$. The three classes are defined by modal dominance [32]: B_1 has $f_2>2f_t$ and $f_2>0.5$ (stationary second-harmonic dominated); B_2 has $f_t>2f_2$ and $f_t>0.5$ (temporally dominated); B_3 satisfies neither condition (mixed). At $Re=500$ and $E^*=0.25$ the representatives used here have $(f_2,f_t)=(0.93,0.07)$ for B_1, $(0.00,1.00)$ for B_2 (seed 5) and $(0.52,0.45)$ for B_3 (seed 0), with a first-harmonic content $f_1=0.006$ for B_3 and below 0.02 for the other two.

The B_1 representative is used in two forms. The *full* series is the control as identified, including its small temporal content; the *mean profile* is its time average, a stationary actuation that isolates the spatial structure. The stationary comparison of Sec. 5 uses the mean profile and the

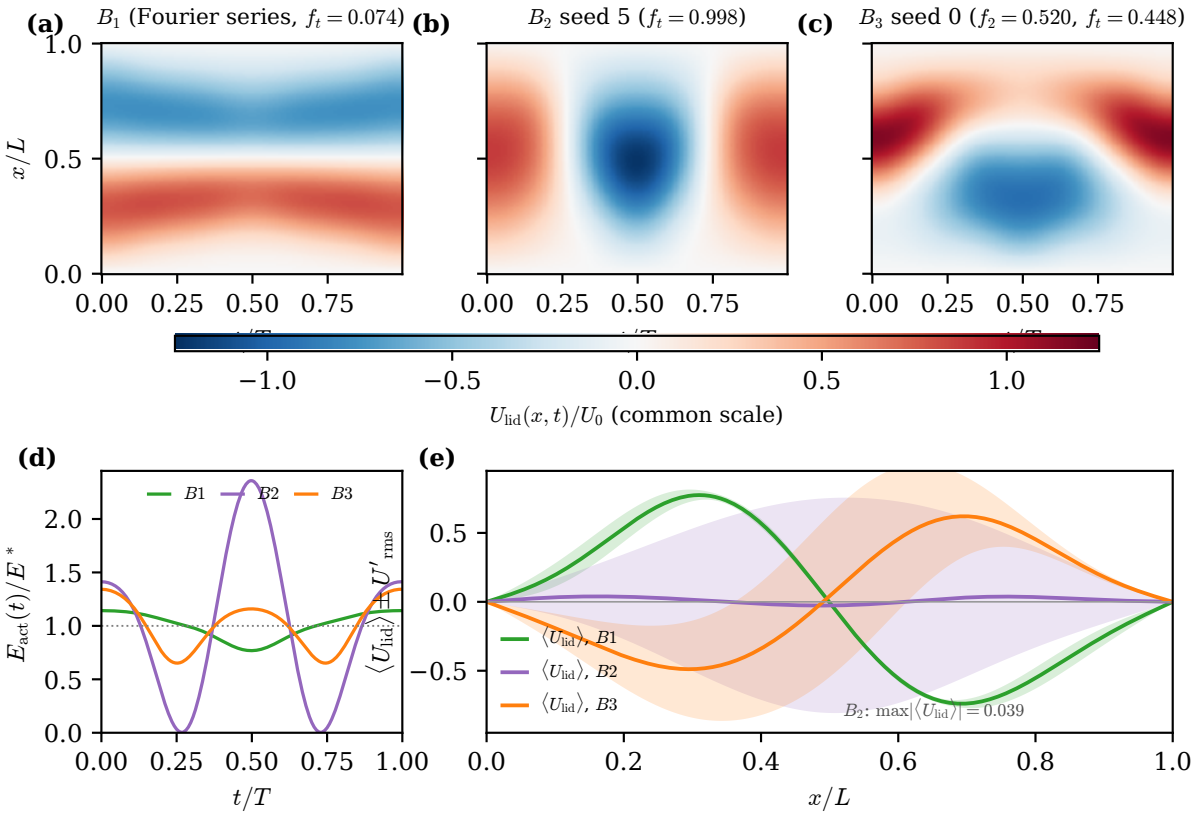


Figure 2: Spatio-temporal structure of the three identified branch controls at $Re = 500$ and $\overline{E}_{\mathrm{act}} = 0.25$. (a)–(c) Lid velocity $U_{\mathrm{lid}}(x,t)$ over one forcing period, on a common colour scale. B_1 is nearly stationary; B_2 oscillates with a full sign reversal, the wall velocity vanishing twice per period; B_3 combines a stationary two-lobe pattern with a strong oscillation. (d) Instantaneous actuation norm $E_{\mathrm{act}}(t)$ normalized by the common cycle-mean target: B_2 spans 0.004–2.36 E^*, whereas B_1 stays within 0.77–1.14 E^* and B_3 within 0.65–1.34 E^*. The comparison is therefore made at equal *cycle-mean* norm, not at equal instantaneous norm. (e) Time-mean lid profile with the temporal standard deviation as a band; the B_2 mean profile reaches only 0.039, so a mean-field diagnostic sees almost no actuation.

cycle-mean comparison of Sec. 6.2 uses the full series, so that the effect of the spatial structure is never conflated with the effect of the temporal one. Representative profiles and their space–time structure are shown in Fig. 2; Table 1 collects the characteristics of the eight primary actuation cases compared in the paper.

4 Numerical methodology

4.1 Identification of the actuation

The control coefficients $a_{mn}(Re)$ and the flow field are obtained simultaneously by a parametric physics-informed neural network with inputs $(x, y, t, \widetilde{Re})$ and a multi-objective loss combining the momentum and continuity residuals, the boundary conditions, a viscous-dissipation objective, the actuation-norm constraint $(E_{\mathrm{act}} - E^*)^2$, a spatial-variance regularizer and a Reynolds-separation term. The identification is performed once over $Re \in [100, 1000]$; the branch representatives at $Re = 500$, $E^* = 0.25$ are then exported on one forcing period and transferred without retraining. The network architecture, the sampling protocol, the loss weights, the training schedule and the branch-classification rules are reported in the Supplementary Material (Secs. S1–S2) and in Ref. [32]; the present paper uses the identified controls as physical boundary data and does not re-examine the identification procedure. What matters here is only that the exported profile is

Table 1: Actuation modes and hydrodynamic states compared in this study ($Re = 500$, fixed viscosity and geometry). $U_{\max}$ is the maximum of the time-mean lid profile and $Re_{\max} = U_{\max} Re$ the corresponding effective Reynolds number; for the time-dependent controls, the maximum over the whole forcing period is given in parentheses. All stationary profiles and the cycle means of the time-dependent ones carry the same quadratic actuation norm, $\overline{E}_{\mathrm{act}} = 0.25$; the instantaneous norm of B_1, B_2 and B_3 varies over the period. f_2 and f_t are the stationary second-harmonic and total temporal energy fractions of the control coefficients. For the pure first-harmonic control, $f_1 = 1$ and $f_2 = 0$; for the pure second-harmonic control, $f_1 = 0$ and $f_2 = 1$.

Actuation	Structure	$U_{\max}$	$Re_{\max}$	f_2	f_t	Role
Uniform lid	constant, sharp corners	0.50	250	—	0	reference
Uniform lid, $\delta = 0.05$	constant, tapered	0.52	258	—	0	converged ref.
$k = 1$	$\sin \pi x$	0.71	354	0.00	0	control
$k = 2$	$\sin 2\pi x$	0.71	354	1.00	0	control
B_1 mean profile	mean of B_1	0.78	390	0.93	0	stationary
B_1 full series	Fourier series	0.78 (0.82)	388 (410)	0.93	0.07	branch
B_2 seed 5	temporal	0.04 (1.23)	20 (615)	0.00	1.00	unsteady
B_3 seed 0	mixed	0.62 (1.16)	311 (580)	0.52	0.45	intermediate

the one the optimizer produced, without harmonic substitution or spatial idealization, and that its realized norm is the prescribed one to within 0.16%.

4.2 Independent MRT-LBM evaluation

The flows are computed with a multiple-relaxation-time D2Q9 lattice-Boltzmann solver [26, 27] that is independent of the identification framework: a Cartesian lattice instead of scattered collocation, a different discretization of the same equations, and a different iteration strategy. Collision is performed in moment space with a diagonal relaxation matrix, and the moving lid is imposed with the Zou–He velocity boundary condition [28, 29]. The solver reproduces the benchmark centerline velocities of Ghia et al. [1] with $L_2(u) = 0.0051$ and $L_\infty(u) = 0.0128$ at $Re = 100$, and $L_2(u) = 0.0141$ and $L_\infty(u) = 0.0300$ at $Re = 1000$ on the 256×256 lattice (Supplementary Sec. S8.2), which fixes the resolution used for the controlled flows.

Each actuation is advanced from rest with the lid switched on at $t = 0$. At least 200 forced periods are discarded, a drift test on the cycle-averaged dissipation fixes the end of the settling phase, and 64 consecutive periods are retained for the statistics. The same protocol is used for every case, so the comparison does not depend on the transient length. Instantaneous dissipation, kinetic energy and lid power are accumulated at every step; the cycle mean is a direct accumulation over the 64 sampled periods, not an eight-point phase quadrature, because the latter is 5% low for the most unsteady responses. The wall shear in Eq. (4) is evaluated with a first-order one-sided derivative of the recovered wall velocity, and the resulting estimator uncertainty is quantified in Sec. 6.4.

4.3 Grid convergence and uncertainty

The three stationary controls and both uniform references are computed on $N = 128$, 256 and 512 lattices. The observed order of convergence p_{obs} and the fine-grid grid-convergence index are obtained by Richardson extrapolation with a safety factor $F_s = 1.25$ [30, 31]. A triple is flagged, and its extrapolated value discarded, when it is non-monotone, when $p_{\mathrm{obs}} \notin [0.1, 4]$, or when the GCI exceeds 100%: in those cases the smooth-error assumption underlying the estimator is not credible. The estimator is applied both to the absolute dissipation and to the dissipation ratios, since the ratios are the quantities used in the physical discussion and their errors need not cancel. The time-dependent branches are ranked on a single $N = 256$ lattice, with two realizations per branch; that restriction is stated wherever those numbers are used.

Table 2: Stationary controls at $Re = 500$ and matched quadratic actuation norm $E^* = 0.25$ on three grids. ε_N is the integrated viscous dissipation; p_{obs} and GCI_{512} are the observed order and the fine-grid grid-convergence index of the absolute dissipation ($F_s = 1.25$). The last column gives the dissipation normalized by the sharp and by the corner-regularized uniform lid on the same grid (the two values are separated by a slash). The B_1 mean profile is used as computed ($E_r = 0.2503$). A dagger marks a flagged triple, for which the extrapolated value is not used. $U_{\max}$ and $Re_{\max}$ are repeated here because the norm constraint fixes the rms wall velocity (0.5) but not the peak.

Control	$U_{\max}$	$Re_{\max}$	ε_{128}	ε_{256}	ε_{512}	p_{obs}	GCI_{512} (%)	$\varepsilon_{512}/\varepsilon_{512}^{\mathrm{ref}}$ (sharp / reg.)
Uniform lid (sharp)	0.50	250	0.00774	0.00948	0.01102	0.18	130.71†	1.000 / 1.272
Uniform lid, $\delta = 0.05$	0.52	258	0.00717	0.00815	0.00866	0.96	7.79	0.786 / 1.000
$k = 1$ ($\sin \pi x$)	0.71	354	0.00635	0.00688	0.00713	1.09	3.85	0.647 / 0.823
$k = 2$ ($\sin 2\pi x$)	0.71	354	0.01012	0.01129	0.01185	1.04	5.62	1.076 / 1.368
B_1 mean profile	0.78	390	0.01067	0.01189	0.01250	1.01	6.01	1.134 / 1.443

5 Energetic response of stationary wall-actuation modes

5.1 Comparison at matched actuation norm

Table 2 and Fig. 3 give the integrated dissipation of the five stationary controls at $Re = 500$ and $E^* = 0.25$. On the finest grid the first harmonic is the only control that lowers the dissipation relative to the uniform lid of the problem statement: $\varepsilon_{512} = 0.00713$ against 0.01102, a ratio of 0.647. The second harmonic raises it to 0.01185 (ratio 1.076) and the B_1 mean profile to 0.01250 (ratio 1.134). The ordering $k = 1 <$ uniform $< k = 2 \approx B_1$ is the central stationary result, and it is the reverse of the fixed-lid-velocity ordering reported in Ref. [32], where both harmonic profiles were less dissipative than the uniform lid and the second harmonic was the least dissipative of the three. The reversal is not a refinement of that result but a consequence of the constraint: at equal lid speed the harmonics have half the norm of the uniform lid, and restoring the norm raises their wall velocity by $\sqrt{2}$ and their effective Reynolds number with it.

The ratios built on the sharp uniform lid are not converged estimates of the shape effect, because the reference itself does not converge: its increments between successive grids decay by only 12% per refinement ($p_{\mathrm{obs}} = 0.18$, GCI 131%), and the formal Richardson limit, 2.25×10^{-2}, is more than twice the finest-grid value and is not a credible continuous-limit dissipation. The three other controls converge, with $p_{\mathrm{obs}} = 1.09$, 1.04 and 1.01 and fine-grid GCIs of 3.9%, 5.6% and 6.0%. Normalizing by the corner-regularized lid instead gives ratios of 0.823, 1.368 and 1.443 at $N = 512$, i.e. -17.7%, $+36.8\%$ and $+44.3\%$, with Richardson estimates 0.802, 1.347 and 1.427 and ratio GCIs of 3.1%, 1.9% and 1.4%. The sign of the ordering is the same under both references and at all three resolutions; only its magnitude changes, because the non-converged corner contribution inflates the denominator of the sharp-lid ratios. We therefore quote the regularized ratios as the better-defined estimate of the shape effect and the sharp-lid ratios as the values directly comparable with the classical uniform-lid baseline, and we do not use the sharp-lid ratio for $k = 1$ as a quantitative reduction: -35.3% overstates the -17.7% obtained against a converged reference. The separation between $k = 2$ and the B_1 mean profile (1.368 against 1.443, about 5%) exceeds their GCIs but not by a wide margin, so they are reported as approximately equal rather than strictly ordered.

5.2 The corner singularity and the reference dependence

The non-convergence of the uniform lid is a property of its corners, not of the uniform profile as such. Its dissipation grows by 16% between $N = 256$ and $N = 512$, the signature of a boundary-layer singularity that the grid cannot resolve. Replacing the discontinuity by the cosine taper of Eq. (10) removes 21% of the dissipation at $N = 512$ and, more importantly, changes the growth between the two finest grids from 16% to 6%, i.e. the expected first-order rate: $p_{\mathrm{obs}} = 0.96$, GCI 7.8%, extrapolated $\varepsilon_\infty = 9.20 \times 10^{-3}$. A continuous lid therefore converges, and the singular

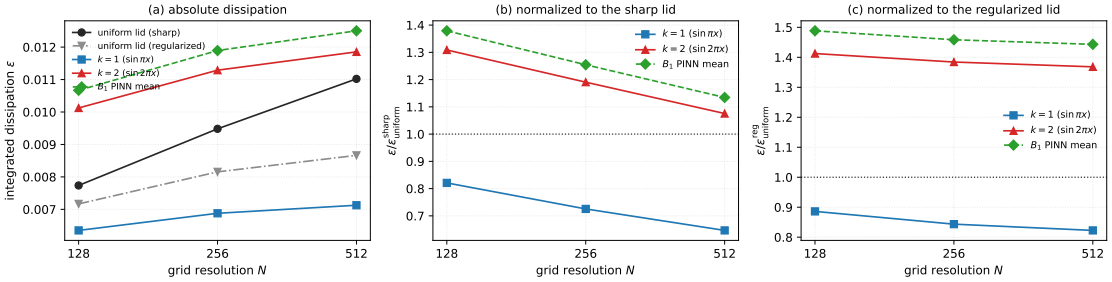


Figure 3: Grid convergence of the stationary comparison at $Re = 500$ and matched actuation norm $E^* = 0.25$. (a) Integrated dissipation versus resolution for the four stationary controls and for the corner-regularized uniform lid. (b) Dissipation normalized by the sharp uniform lid on the same grid: this reference does not converge, and the ratios drift by 18–21% over the triple. (c) Dissipation normalized by the regularized lid: the three ratio triples are monotone and converge with GCIs of 1.4–3.1%. The sign of the ordering $k = 1 <$ uniform $< k = 2 \approx B_1$ is observed at every resolution in both (b) and (c); its magnitude is reference-dependent.

one does not. This has two consequences for the interpretation. First, a comparison of actuation shapes that uses the discontinuous lid as denominator mixes the shape effect with a resolution artefact; the regularized reference is the only one that separates them. Second, the corner is an energetic feature in its own right: the uniform lid concentrates a large part of its dissipation in two small regions whose contribution grows with resolution, so any ranking that depends on the corner peak is resolution-dependent by construction.

The practical conclusion is that two numbers must be reported for each profile, not one. The sharp-lid ratio answers the question "what does this actuation do relative to the classical cavity?" and is the quantity a reader will compare with the literature. The regularized ratio answers "what is the effect of the spatial shape at fixed actuation norm?" and is the quantity that supports a physical statement about shape. Both are given in Table 2, and the text uses the second for every claim about ordering or magnitude.

5.3 Local mechanism of the second-harmonic penalty

The ranking has a local origin that can be read directly from the fields (Fig. 4). The uniform lid concentrates its dissipation in the two upper corners, where the imposed tangential velocity jumps to zero: the peak strain-rate magnitude there is 169 in units of U_0/L, against 15.0 for $k = 1$ and 23.4 for $k = 2$, and those corner layers are the brightest features of the dissipation field. Both harmonics remove the jump, and their dissipation is spread over the lid shear layer and the side walls instead. But the second harmonic pays for this in the interior. Its lid generates a pair of counter-rotating cells of nearly equal strength, separated by a wall-to-wall separatrix at $x = 0.5$, each enclosing its own low-strain core; the uniform and first-harmonic lids recover the classical single primary vortex with a corner eddy. The reorganization shifts the strain-rate distribution towards larger values: the interior mean $\langle |\mathbf{S}| \rangle$ is 1.450 for $k = 2$, against 1.039 for the uniform lid and 1.088 for $k = 1$, a 40% increase at identical actuation norm.

The trade-off is explicit. The first harmonic removes almost all of the corner peak (169 to 15.0) at a cost of 4.7% in interior mean strain, and is a net reduction. The second harmonic removes slightly less of it (23.4) while raising the interior mean by 39.6%, and is a net penalty: it buys a symmetric two-cell state with a stronger interior strain field, and the corner singularity it removes is not costly enough to compensate. Relative to the uniform lid, the increase in interior mean strain is only 4.7% for $k = 1$, but reaches 39.6% for $k = 2$. The latter increase is approximately 8.4 times larger, even though the second harmonic does not reduce the peak strain below that of the first harmonic (23.4 against 15.0). Additional identification-sensitivity and control-representation studies, including loss-term ablation, network-capacity sensitivity, control-space truncation, aspect-ratio dependence and Fourier versus modified-Chebyshev parametrizations,

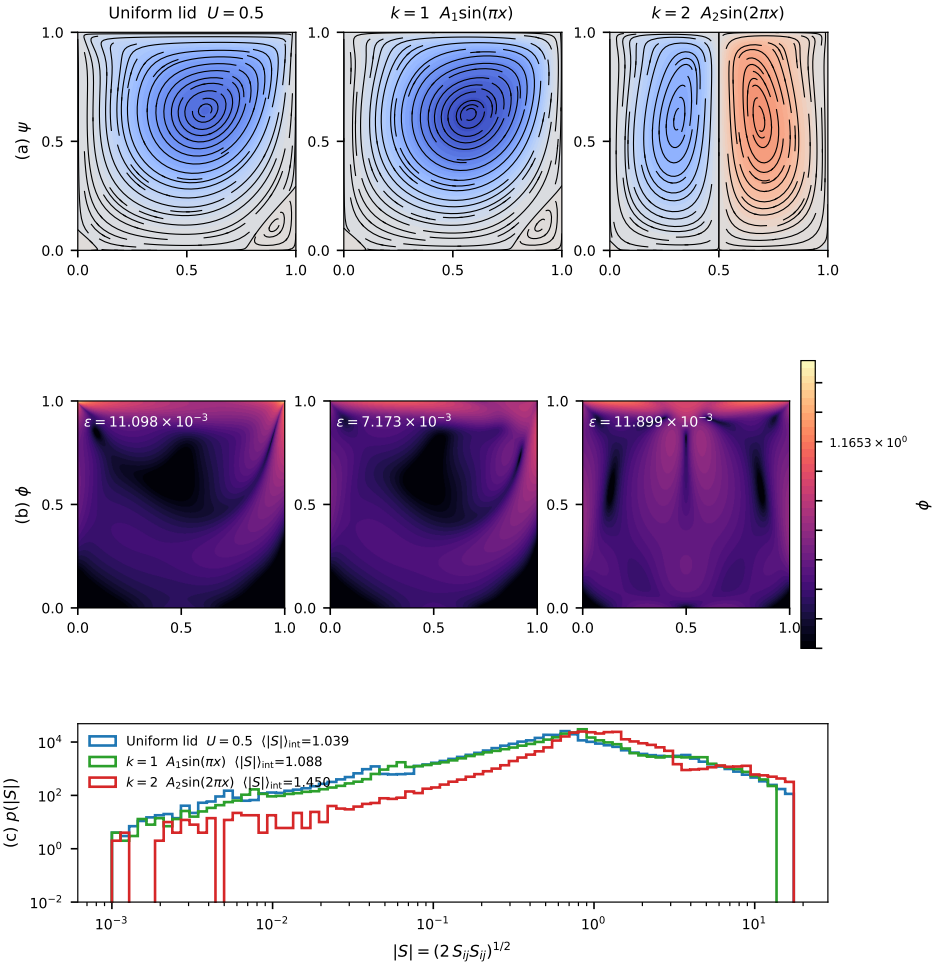


Figure 4: Local origin of the stationary ranking at $Re = 500$, $N = 512$ and matched actuation norm, on scales common to the three columns. (a) Stream function of the steady response with the streamline topology. (b) Local viscous dissipation $\phi = \nu(2S_{11}^2 + 2S_{22}^2 + 4S_{12}^2)$ on a logarithmic scale spanning six decades below the global maximum; the annotation gives the domain integral. (c) Probability density of the strain-rate magnitude over the cavity interior (three boundary cells excluded), with the interior mean $\langle|\mathbf{S}|\rangle_{\rm int}$. The uniform lid peaks at the corners ($|\mathbf{S}|_{\max} = 169$); $k = 1$ removes the peak at almost no interior cost ($\langle|\mathbf{S}|\rangle_{\rm int} = 1.088$ against 1.039); $k = 2$ removes less of it (23.4) and raises the interior mean to 1.450, which is the origin of its dissipation penalty.

are documented in Supplementary Secs. S3–S7, Tables S3–S11 and Figs. S1–S4. These studies characterize the robustness of branch identification; the dissipation ordering discussed here is established independently by the MRT-LBM calculations and their grid and estimator checks in Supplementary Secs. S8.4–S8.7.

6 Cycle-mean dissipation of time-dependent actuation branches

6.1 Why the dissipation of the mean flow is not the cycle mean

For the stationary controls of Sec. 5 the distinction between $D(\overline{\mathbf{u}})$ and $\overline{D}$ is empty, because the response is steady. For the time-dependent branches it becomes the central issue. Equation (7) is exact: the cycle mean exceeds the mean-field value by the dissipation of the fluctuations, and there is no reason for that term to be small when the actuation reverses sign. A mean-field diagnostic answers the question "how much would this flow dissipate if it were steady?"; it does not answer

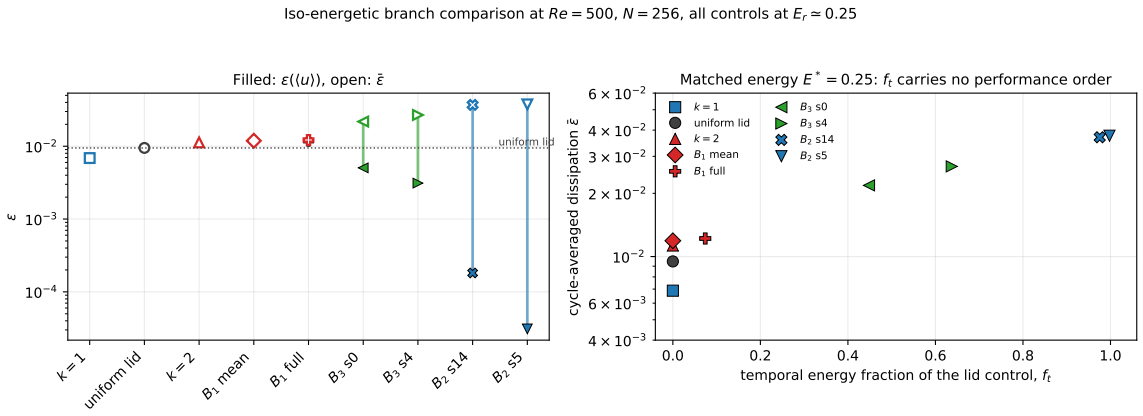


Figure 5: Iso-energetic comparison of the nine controls at $Re = 500$, $\overline{E}_{\mathrm{act}} = 0.25$ and $N = 256$. Left: the two dissipation functionals of each control, the dissipation of the time-mean field (filled markers) and the cycle mean (open markers), joined by the gap between them; the two coincide for stationary controls. Right: cycle-mean dissipation against the temporal energy fraction f_t of the lid control. The ordering in f_t is the order in which the branches are recovered by the identification, and it is not the order of the dissipation cost.

"how much does it dissipate?". Figure 5 and Table 3 show how far apart the two answers can be; supplementary Tables S15–S16 provide additional cycle-averaged statistics and iso-energetic comparisons, while Figs. S5–S6 provide the phase-resolved field captures.

6.2 Cycle-mean comparison of the three branches

Table 3 collects the result. The five quasi-stationary controls span less than a factor of two, from 6.88×10^{-3} for $k = 1$ to 1.22×10^{-2} for the full B_1 series, and their mean-field and cycle-mean values coincide to within 4%; for the B_1 series the difference is 1.17×10^{-2} against 1.22×10^{-2}, which is the residual effect of its small temporal content. The branches with substantial temporal content leave that band. B_2 dissipates 3.76×10^{-2} over the cycle while its mean field dissipates 3.10×10^{-5}, a ratio of 1214; its mean field holds only 6% of the cycle-mean kinetic energy, so the mean motion is a shallow recirculation under the lid, not a weakened copy of the instantaneous flow. B_3 is intermediate: $D(\overline{\mathbf{u}}) = 5.05 \times 10^{-3}$ and $\overline{D} = 2.18 \times 10^{-2}$, a ratio of 4.3. On the cycle average, and at the same actuation norm, B_2 costs 3.1 times B_1 and 5.5 times the least dissipative stationary control; its two realizations agree to 1.7%, whereas the two B_3 realizations differ by 23%, so the B_3 value quoted here is the lower of the two and must be read as a lower bound on that class.

Three statements follow, and they are the ones the paper retains. First, the mean-field functional is not a surrogate for the cycle mean: it can be three orders of magnitude too small and it reverses the ranking of the branches. Second, the temporal fraction f_t orders the branches by accessibility, not by cost: B_2, the most strongly time-dependent control, is the most dissipative of the three at matched norm. Third, the ranking of the time-dependent branches ($B_1 < B_3 < B_2$) is the same as the ranking of their mean-field dissipation read in reverse, which is a direct consequence of Eq. (7): the branch that suppresses its mean flow most effectively is the one that pays most in fluctuations.

6.3 Flow organization behind the cycle-mean cost

The field-level organization explains the numbers (Figs. 6 and 7). The B_1 time-mean flow is the familiar pair of counter-rotating cells centred at $(0.31, 0.54)$ and $(0.69, 0.56)$, with $\max|\langle \mathbf{u} \rangle| = 0.733$ and $\langle \psi \rangle \in [-0.036, 0.035]$; half of its mean kinetic energy lies below $y = 0.675$, so the cells fill the cavity. Its instantaneous states coincide with that mean to 2.7% in field norm and its kinetic-energy

Table 3: Dissipation of the time-mean field and cycle-mean dissipation at $Re = 500$, $N = 256$ and cycle-mean actuation norm $\overline{E}_{\mathrm{act}} = 0.25$. $D(\overline{\mathbf{u}})$ is the functional evaluated on the time-mean field; $\overline{D}$ is the average of the instantaneous dissipation over the phase-resolved periodic response and is the performance index relevant to a periodic flow. The two coincide only in the steady limit. A_ε is the half-spread of $D(t)$ over the cycle, i.e. the variability concealed by a single cycle-mean number. The B_1 mean profile row is stationary by construction; the B_1 full row is the identified series.

Control	f_t	$\overline{E}_{\mathrm{act}}$	$D(\overline{\mathbf{u}})$	$\overline{D}$	$\overline{D}/D(\overline{\mathbf{u}})$	A_ε
Uniform lid	0.000	0.250	9.48×10^{-3}	9.48×10^{-3}	1.00	0
$k = 1$ ($\sin \pi x$)	0.000	0.250	6.88×10^{-3}	6.88×10^{-3}	1.00	0
$k = 2$ ($\sin 2\pi x$)	0.000	0.250	1.13×10^{-2}	1.13×10^{-2}	1.00	0
B_1 mean profile	0.000	0.250	1.19×10^{-2}	1.19×10^{-2}	1.00	0
B_1 full series	0.074	0.250	1.17×10^{-2}	1.22×10^{-2}	1.04	2.4×10^{-3}
B_2 seed 5	0.998	0.250	3.10×10^{-5}	3.76×10^{-2}	1214	3.6×10^{-2}
B_2 seed 14	0.975	0.250	1.82×10^{-4}	3.70×10^{-2}	204	3.5×10^{-2}
B_3 seed 0	0.448	0.250	5.05×10^{-3}	2.18×10^{-2}	4.3	1.5×10^{-2}
B_3 seed 4	0.637	0.250	3.12×10^{-3}	2.69×10^{-2}	8.6	2.6×10^{-2}

Table 4: Mechanical energy budget of the three branch controls at $Re = 500$ and $N = 256$, at the common cycle-mean actuation norm $E^* = 0.25$: B_1 is the full series ($f_t = 0.074$), B_2 is seed 5 ($f_t = 0.998$) and B_3 is seed 0 ($f_2 = 0.520$, $f_t = 0.448$). All quantities are cycle averages over 64 forced periods of the instantaneous fields; $\overline{D}$ is never replaced by $D(\overline{\mathbf{u}})$. $\sigma(\overline{D})$ is the block standard error over eight blocks of eight cycles, expanded by $t_{7,0.975} = 2.365$; it is below 0.1% for all three branches because the response is periodic to better than 10^{-3}, so the physically relevant uncertainty is the seed-to-seed spread (1.7% within B_2, 23% within B_3). The residual $\overline{P}_{\mathrm{lid}} - \overline{D}$ is the first-order truncation of the wall-shear estimator, not a physical imbalance (Sec. 6.4).

Branch	$\overline{E}_{\mathrm{act}}$	$\overline{K}$	$\overline{P}_{\mathrm{lid}}$	$\overline{D}$	$\sigma(\overline{D})$
B_1 (full series)	0.2500	1.15×10^{-2}	1.18×10^{-2}	1.22×10^{-2}	$< 0.1\%$
B_2 (seed 5)	0.2504	2.47×10^{-4}	3.76×10^{-2}	3.76×10^{-2}	$< 0.1\%$
B_3 (seed 0)	0.2503	3.98×10^{-3}	2.16×10^{-2}	2.18×10^{-2}	$< 0.1\%$

fluctuation ratio is $R_K = 4 \times 10^{-4}$, so it is quasi-stationary in the operational sense. The B_2 mean field is 19 times weaker in stream function ($\max |\langle \psi \rangle| = 0.0019$) and 18 times weaker in peak speed (0.042, or 5.7% of the B_1 value); its mean extrema sit at $y = 0.89$–0.92 and half of the mean kinetic energy lies above $y = 0.95$. Its streamwise fluctuation, by contrast, reaches 0.555—thirteen times its own mean and 76% of the B_1 mean peak. The B_3 flow keeps an intermediate mean circulation ($\max |\langle \mathbf{u} \rangle| = 0.595$, 81% of B_1) but places it close to the lid, with extrema at $(0.20, 0.86)$ and $(0.77, 0.81)$ and half of the mean kinetic energy above $y = 0.91$; its fluctuation reaches 0.416.

"Weak mean flow" is therefore not "weak flow". The controls with a strong temporal component do not produce a small perturbation around a mean recirculation: they produce responses whose fluctuations dominate the instantaneous field while the mean is nearly zero. A figure that normalized each panel by its own maximum, or a diagnostic that used only the mean field, would hide exactly the contrast that defines the two unsteady branches.

6.4 Mechanical energy budget of the branches

Table 4 and Fig. 8 close the comparison with the budget of Eq. (8). The kinetic energy separates the branches by two orders of magnitude (1.15×10^{-2}, 2.47×10^{-4} and 3.98×10^{-3}), while the cycle-mean dissipation orders them the other way round (1.22×10^{-2}, 3.76×10^{-2} and 2.18×10^{-2}). Within the period, $D(t)$ varies by a factor 1.5, 10.9 and 4.2 respectively, and the lid power of B_2 is negative for 14% of the period: the flow returns energy through the lid while the Stokes layer spins down. The residual $\overline{P}_{\mathrm{lid}} - \overline{D}$, computed with the wall velocity recovered at the boundary node rather than the prescribed profile (Sec. S8.7), is -2.9%, -0.1% and -0.9%.

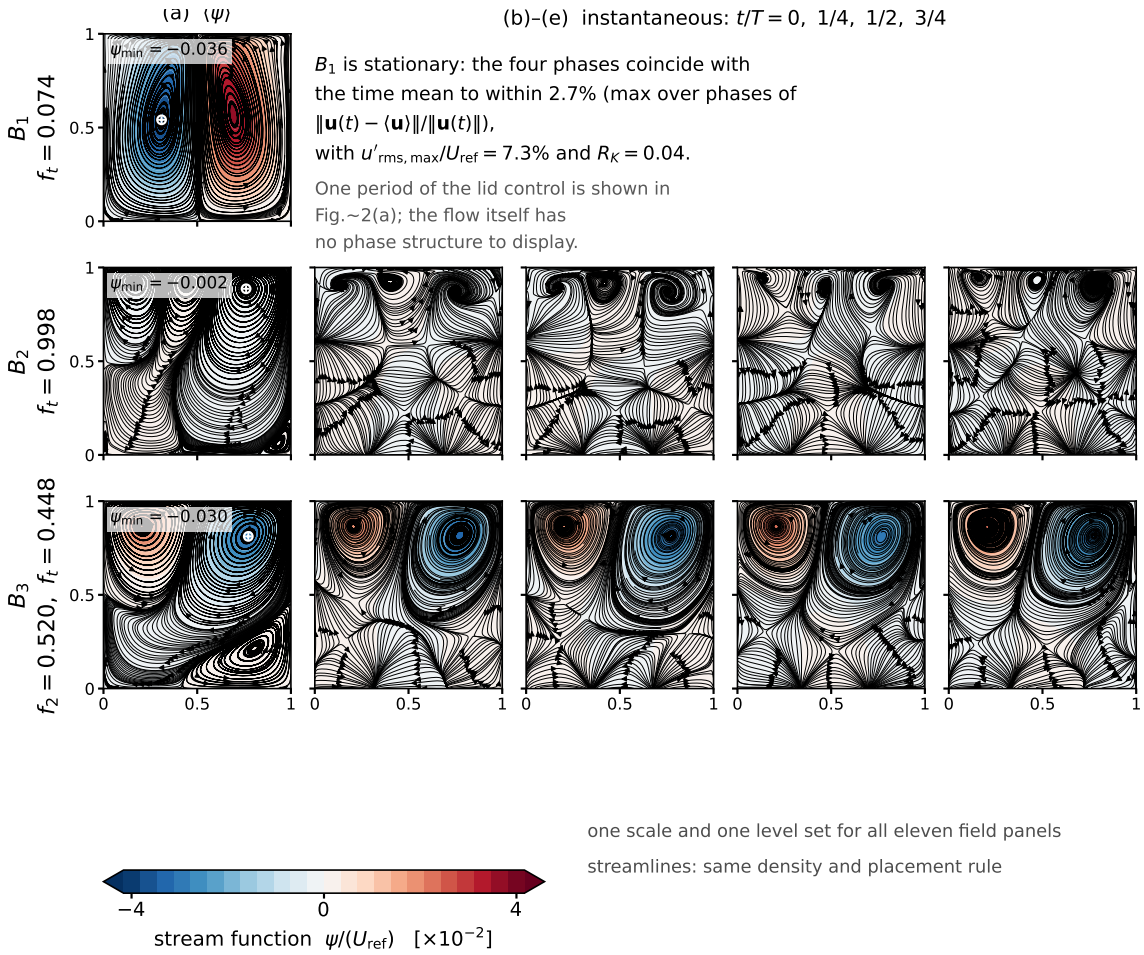


Figure 6: Flow topology of the three branch controls at $Re = 500$, $E^* = 0.25$ and $N = 256$, on one stream-function scale and one set of contour levels common to all panels. Rows: B_1 (full series, $f_t = 0.074$), B_2 (seed 5, $f_t = 0.998$) and B_3 (seed 0, $f_2 = 0.520$, $f_t = 0.448$). (a) Time mean $\langle\psi\rangle$ with the streamlines of the mean flow; the circle marks its negative extremum. (b)–(e) Instantaneous ψ and streamlines at $t/T = 0, 1/4, 1/2, 3/4$ for B_2 and B_3; the four phases belong to the same periodic state, whose first- and second-half phase averages differ by at most 1.0×10^{-3} in relative field norm. For B_1 the instantaneous states coincide with the mean to 2.7%, so the four panels are replaced by that statement. The common scale is deliberate: the B_2 mean cell is 19 times weaker than the B_1 one and is nearly invisible, which is the point.

The residual is two to three orders of magnitude larger than the drift of K over the sampling window, which is at most $1.5 \times 10^{-2}\%$ of $\overline{D}$; it is therefore not a failure of periodicity. It is the truncation error of the one-sided wall-shear estimate. Two checks bound that error. First, the first-order estimator reproduces the independently accumulated P_{lid} of the same fields to 0.3% for B_1, so it is internally consistent; the residual is a property of the estimator, not of the accumulation. Second, replacing it by a second-order one-sided derivative on the same fields moves the B_1 mean-field residual from -2.87% to $+0.58\%$, i.e. it crosses zero, which shows that the estimator uncertainty is of the order of the residual itself (a few per cent). Applied to the branch cycle means, the same change moves the B_1 residual to $+1.4\%$ but inflates the B_2 and B_3 residuals, because the three-point stencil amplifies lattice-scale noise in the thin Stokes layer. We therefore retain the first-order estimator for all three branches, report the residual as measured, and present Eq. (9) as an energetic verification with a few-per-cent estimator uncertainty rather than as an exact closure. This is sufficient for the conclusions drawn here, which rest on differences of a factor of three, not on the third significant digit.

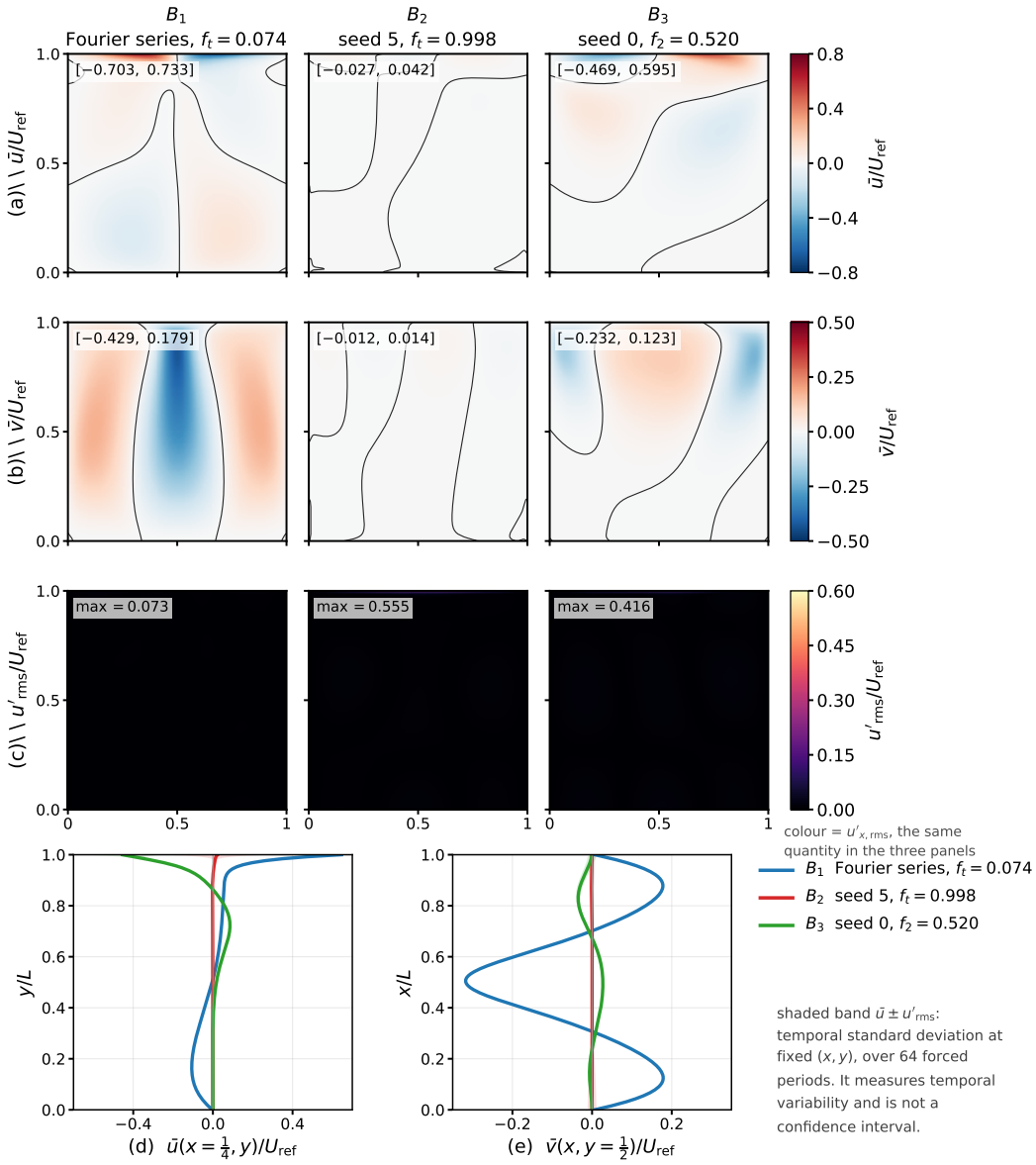


Figure 7: Velocity fields and centreline profiles of the same three captures. Rows: (a) $\bar{u}$, (b) $\bar{v}$ and (c) $u'_{x,\mathrm{rms}}$, each on one colour range for the three columns and a single normalization, with the occupied range written in every panel so that the common scale can be checked. Grey lines are the zero contours of the mean fields. (d) $\bar{u}$ along $x = 1/4$ with the $\pm u'_{\mathrm{rms}}$ band; the midline $x = 1/2$ is not used because the antisymmetric controls have $u \equiv 0$ there. (e) $\bar{v}$ along $y = 1/2$, with $\bar{v}$ on the abscissa. The bands are temporal standard deviations over the sixty-four sampled periods, not confidence intervals.

7 Physical interpretation and implications for flow control

7.1 Why the stationary second harmonic increases dissipation

At matched quadratic actuation norm the second harmonic is not a low-dissipation mode, and the reason is geometric rather than spectral. The lid profile $A_2 \sin(2\pi x)$ imposes two opposite lobes of wall velocity, which drive two counter-rotating cells of comparable strength instead of one primary vortex. That reorganization has two competing energetic consequences. It removes the corner velocity jump, which suppresses a singular strain peak that dominates the dissipation of the uniform lid. It also forces the return flow through a wall-to-wall separatrix at the cavity centre, which raises the strain over the whole interior: the interior mean of $|\mathbf{S}|$ increases by 40% at identical norm. The second effect wins, because the corner layer is thin while the interior is

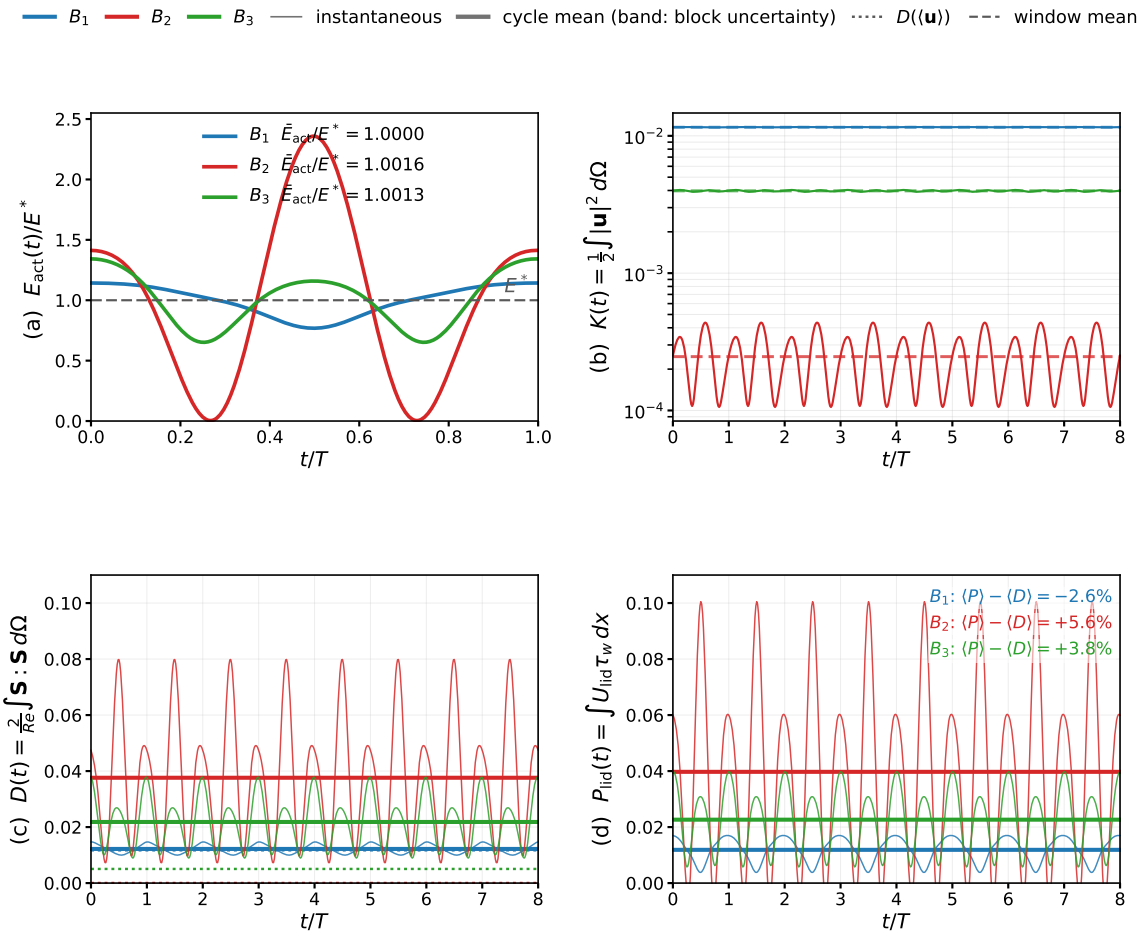


Figure 8: Phase-resolved energetic response of the three branches at $Re = 500$ and $E^* = 0.25$. (a) Instantaneous actuation norm $E_{\mathrm{act}}(t)$ over one forced period, normalized by the common cycle-mean target. (b) Kinetic energy $K(t)$ over the first eight of the sixty-four sampled periods (logarithmic ordinate). (c) Viscous dissipation $D(t)$: instantaneous (thin), cycle mean (thick, with the block uncertainty as a band) and, dotted, the dissipation $D(\langle\mathbf{u}\rangle)$ of the time-mean field, which is not substituted for the cycle mean. (d) Lid power $P_{\mathrm{lid}}(t)$, positive into the fluid, with the same conventions; the annotations give the window-averaged residual $\overline{P}_{\mathrm{lid}} - \overline{D}$ computed with the recovered wall velocity (Sec. S8.7). The kinetic-energy balance integrates to zero over one period of a periodic state, so the two cycle means must agree; the measured residual is the wall-shear estimator error discussed in the text.

not. The first harmonic removes the same corner peak at almost no interior cost, and is the only stationary profile that is a net reduction. The energetic cost of a spatial actuation mode is thus controlled by the volume over which it raises the strain, not by the peak it removes; a mode that is "smoother" at the wall need not be less dissipative in the bulk.

7.2 Why temporal forcing increases the cycle-mean dissipation

For a periodic control the relevant functional is the cycle mean of a quadratic quantity, and the fluctuations are a first-order contribution to it, not a correction. Equation (7) makes this explicit, and the B_2 branch is its clearest illustration: a mean field that dissipates 3.1×10^{-5} and a cycle mean that dissipates 3.8×10^{-2}, with the mean field holding 6% of the kinetic energy. The physical reason is that the oscillating lid generates a Stokes layer whose wall-normal gradients are large and reverse twice per period; the instantaneous strain is therefore large wherever the instantaneous velocity is large, and the time average of its square cannot be recovered from the average velocity. The same mechanism explains why the lid power of B_2 is negative for 14% of the

period: while the Stokes layer spins down, the flow pushes the wall and returns part of the energy that it received earlier. A control designer who evaluates such a state from its mean field will conclude that it is almost frictionless, whereas the cycle average ranks it as the most dissipative of the three branches.

7.3 What branch accessibility does and does not imply

The three branches were not selected by their dissipation. They were selected by a physics-informed optimization whose loss contains a dissipation objective, an actuation-norm constraint and a spatial-variance regularizer, and whose outcome depends on the initialization and on the regularization weight [32]. At $Re = 500$ and $E^* = 0.25$ the stationary second-harmonic structure is the most frequently recovered one, and the full B_1 series is quasi-stationary and reproducible; that is a statement about the variational landscape, and it is confirmed here by the independent solver. It is not a statement about efficiency. At the same actuation norm the branch most frequently recovered is a net dissipation penalty relative to the first harmonic, and the branch with the strongest temporal content, which is also recovered in a reproducible way, is the most dissipative of the three. The operational conclusion is that accessibility and physical performance are different properties, and that a control study which reports only the recovered mode, or only its mean-field dissipation, can invert the energetic ranking of the actuations it compares.

Two limits of this statement must be stated explicitly, because they define its scope. First, the temporal comparison covers three representative realizations at a single Reynolds number ($Re = 500$) and a single grid ($N = 256$), with two realizations for B_2 and B_3; the within-class spread is 1.7% for B_2 but 23% for B_3, so the B_3 cost is a lower bound on its class. The stationary comparison, by contrast, rests on three grids and two reference lids. Second, the B_1 temporal content is small but not zero ($f_t = 0.074$), and its cycle mean exceeds its mean-field value by 4%; the quasi-stationary limit is approached, not attained. Generalization to other Reynolds numbers, to three-dimensional cavities, or to actuations with a different spatial support is not implied by the present results.

8 Conclusions

Four statements summarize the study.

1. *The spatial structure of wall actuation reorganizes the cavity and the strain field.* At matched quadratic actuation norm the second spatial harmonic replaces the single primary vortex of the uniform lid by two counter-rotating cells and raises the interior mean strain rate by 40%; the first harmonic removes the corner strain peak at almost no interior cost. The corner singularity of the discontinuous lid is an energetic feature in its own right: it prevents the convergence of the reference dissipation ($p_{\mathrm{obs}} = 0.18$, GCI 131%) and disappears once the corner is regularized ($p_{\mathrm{obs}} = 0.96$, GCI 7.8%).

2. *The ranking of stationary profiles depends on the constraint.* At equal actuation norm, and relative to a converged corner-regularized reference, the first harmonic reduces the dissipation by 17.7%, while the second harmonic and the mean profile of the identified stationary branch increase it by 36.8% and 44.3%. At equal lid speed the same three profiles are all less dissipative than the uniform lid and the second harmonic is the best of them [32]; the inversion is produced by the norm constraint, which fixes the rms wall velocity but not the peak, and therefore not the effective Reynolds number ($Re_{\max} = 250$ for the uniform lid, 354 for the harmonics, 390 for the B_1 mean profile at $E^* = 0.25$).

3. *The cycle-mean dissipation of a time-dependent actuation is not the dissipation of its mean field.* For the temporally dominated branch the two differ by a factor 1214, and the branch that is most strongly damped in the mean is the most dissipative on the cycle (3.76×10^{-2},

i.e. 3.1 times the quasi-stationary branch and 5.5 times the first harmonic). The fluctuating part of the strain field carries the cost, and Eq. (7) makes that statement exact.

4. *The branch recovered by optimization is not the branch that minimizes dissipation at fixed control effort.* The most frequently recovered stationary structure is a net penalty at matched norm, and the most strongly time-dependent branch is the most dissipative of the three. Accessibility, reproducibility and physical performance are distinct properties of an actuation, and only the last is measured by the cycle-mean dissipation.

The practical implication for modal flow control is that a comparison of actuation strategies must state its constraint explicitly. Equal wall speed, equal quadratic norm and equal mechanical power define three different problems and can produce three different rankings; the present study uses the quadratic norm because it is the constraint available to an optimizer, and reports the peak velocity so that the effective Reynolds number of each comparison is visible. The cycle-mean dissipation, not the mean-field value and not the modal spectrum, is the performance index that follows from the energy identity.

Three extensions follow directly. A grid study of the time-dependent branches would establish whether the factor-of-three separation is resolution-independent to the same degree as the stationary ranking. Retraining under an explicit cycle-mean norm constraint, rather than rescaling an identified control after the fact, would separate the effect of the constraint from the effect of the identification path. Finally, three-dimensional cavities would introduce spanwise modes and test whether the two-cell second-harmonic state survives when the flow can vary along the span. The physics-informed identification remains, in this programme, the tool that generates the candidate actuations; the fluid-mechanical evaluation is what decides between them.

Data availability

The actuation series, the field captures and the post-processing scripts that support the figures and tables of this article are available at https://github.com/Ahmedalgeria778/PoF_cavity_dissipation_study and from the corresponding author on reasonable request.

Acknowledgments

The authors thank the Laboratory of Fluid Mechanics of the Ecole Militaire Polytechnique for computational support.

References

[1] U. Ghia, K. N. Ghia, and C. T. Shin, "High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method," J. Comput. Phys. **48**, 387–411 (1982).

[2] O. Botella and R. Peyret, "Benchmark spectral results on the lid-driven cavity flow," Comput. Fluids **27**, 421–433 (1998).

[3] P. N. Shankar and M. D. Deshpande, "Fluid mechanics in the driven cavity," Annu. Rev. Fluid Mech. **32**, 93–136 (2000).

[4] S. Albensoeder and H. C. Kuhlmann, "Accurate three-dimensional lid-driven cavity flow," J. Comput. Phys. **206**, 536–558 (2005).

[5] H. C. Kuhlmann and F. Romanò, "The lid-driven cavity," in *Computational Modelling of Bifurcations and Instabilities in Fluid Dynamics*, edited by A. Gelfgat (Springer, Cham, 2019), pp. 233–309.

[6] M. Gad-el-Hak, *Flow Control: Passive, Active, and Reactive Flow Management* (Cambridge University Press, Cambridge, 2000).

[7] S. L. Brunton and B. R. Noack, "Closed-loop turbulence control: Progress and challenges," Appl. Mech. Rev. **67**, 050801 (2015).

[8] C. W. Rowley and S. T. M. Dawson, "Model reduction for flow analysis and control," Annu. Rev. Fluid Mech. **49**, 387–417 (2017).

[9] T. R. Bewley, P. Moin, and R. Temam, "DNS-based predictive control of turbulence: an optimal benchmark for feedback algorithms," J. Fluid Mech. **447**, 179–225 (2001).

[10] M. Bergmann, L. Cordier, and J.-P. Brancher, "Optimal rotary control of the cylinder wake using proper orthogonal decomposition reduced-order model," Phys. Fluids **17**, 097101 (2005).

[11] H. Choi, P. Moin, and J. Kim, "Active turbulence control for drag reduction in wall-bounded flows," J. Fluid Mech. **262**, 75–110 (1994).

[12] M. Quadrio and P. Ricco, "Critical assessment of turbulent drag reduction through spanwise wall oscillations," J. Fluid Mech. **521**, 251–262 (2004).

[13] T. C. Corke, C. L. Enloe, and S. P. Wilkinson, "Dielectric barrier discharge plasma actuators for flow control," Annu. Rev. Fluid Mech. **42**, 505–529 (2010).

[14] J. Rabault, M. Kuchta, A. Jensen, U. Réglade, and N. Cerardi, "Artificial neural networks trained through deep reinforcement learning discover control strategies for active flow control," J. Fluid Mech. **865**, 281–302 (2019).

[15] P. Garnier, J. Viquerat, J. Rabault, A. Larcher, A. Kuhn, and E. Hachem, "A review on deep reinforcement learning for fluid mechanics," Comput. Fluids **225**, 104973 (2021).

[16] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," J. Comput. Phys. **378**, 686–707 (2019).

[17] S. Cai, Z. Mao, Z. Wang, M. Yin, and G. E. Karniadakis, "Physics-informed neural networks (PINNs) for fluid mechanics: A review," Acta Mech. Sin. **37**, 1727–1738 (2021).

[18] B. R. Noack, K. Afanasiev, M. Morzyński, G. Tadmor, and F. Thiele, "A hierarchy of low-dimensional models for the transient and post-transient cylinder wake," J. Fluid Mech. **497**, 335–363 (2003).

[19] P. J. Schmid, "Dynamic mode decomposition of numerical and experimental data," J. Fluid Mech. **656**, 5–28 (2010).

[20] K. Taira, C. W. Rowley, O. T. Schmidt, D. S. Luchtenburg, S. L. Brunton, and B. R. Noack, "Modal analysis of fluid flows: An overview," AIAA J. **55**, 4013–4041 (2017).

[21] D. Sipp and A. Lebedev, "Global stability of base and mean flows: a general approach and its applications to cylinder and open cavity flows," J. Fluid Mech. **593**, 333–358 (2007).

[22] S. Beneddine, D. Sipp, A. Arnault, J. Dandois, and F. Lesshafft, "Conditions for validity of mean flow stability analysis," J. Fluid Mech. **798**, 485–504 (2016).

[23] S. B. Pope, *Turbulent Flows* (Cambridge University Press, Cambridge, 2000).

[24] L. D. Landau and E. M. Lifshitz, *Fluid Mechanics*, 2nd ed. (Pergamon, Oxford, 1987).

[25] C. R. Doering and C. Foias, "Energy dissipation in body-forced turbulence," J. Fluid Mech. **467**, 289–306 (2002).

[26] D. d'Humières, "Multiple-relaxation-time lattice Boltzmann models in three dimensions," Philos. Trans. R. Soc. Lond. A **360**, 437–451 (2002).

[27] S. Chen and G. D. Doolen, "Lattice Boltzmann method for fluid flows," Annu. Rev. Fluid Mech. **30**, 329–364 (1998).

[28] Q. Zou and X. He, "On pressure and velocity boundary conditions for the lattice Boltzmann BGK model," Phys. Fluids **9**, 1591–1598 (1997).

[29] X. F. Pan, A. Xu, G. Zhang, and S. Jiang, "Lattice Boltzmann simulation of particle-laden turbulent channel flow," Comput. Fluids **35**, 1432–1444 (2006).

[30] P. J. Roache, "Perspective: A method for uniform reporting of grid refinement studies," J. Fluids Eng. **116**, 405–413 (1994).

[31] I. B. Celik, U. Ghia, P. J. Roache, and C. J. Freitas, "Procedure for estimation and reporting of uncertainty due to discretization in CFD applications," J. Fluids Eng. **130**, 078001 (2008).

[32] A. Beniaiche, "Physics-informed identification and independent numerical verification of conditional modal collapse in active control of lid-driven cavity flow," Phys. Fluids **38**, 103612 (2026). doi:10.1063/5.0351347